Unit – 5

Singular Value Decomposition

Question Bank

1. If $Q(x) = x^2 - 8xy - 5y^2$ then compute Q(-3,1)
2. Find the singular values of A = $\begin{pmatrix} 9 & -9 \\ -9 & 9 \end{pmatrix}$
3. Test for positive definiteness A = $\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$
4. Find the SVD of A= $\begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$
5. Find The Quadratic form $A = \begin{pmatrix} -2 & 1 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$
6. Find The Quadratic form $A = \begin{pmatrix} -2 & 2 \\ 2 & 2 \end{pmatrix}$
7. Find the matrix of the Quadratic form 5x2 + $6y^2$ + $7z^2$ + 4xy - 4yz
8. Find the Quadratic form of $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$
9. Find the singular values of $\begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}$
10. Find SVD of $A = \begin{pmatrix} \sqrt{3} & 2 \\ 0 & \sqrt{3} \end{pmatrix}$
11. Diagonalize A = $\begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & 1 \\ 3 & 1 & 1 \end{pmatrix}$
12. Find the singular value decomposition of A = $\begin{pmatrix} 7 & 1 \\ 0 & 0 \\ 5 & 5 \end{pmatrix}$
13. Find the eigen values of A = $\begin{pmatrix} 4 & -2 \\ 2 & -1 \\ 0 & 0 \end{pmatrix}$
14. Define eigen space and find the eigen space of the eigen values of $\begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & 1 \\ 3 & 1 & 1 \end{pmatrix}$